OGC® DOCUMENT: DEMO-LEVELS

Open Geospatial Consortium

STATEMENT LEVELS EXPLAINED: REQUIREMENT, RECOMMENDATION, PERMISSION

DRAFT STANDARD
Implementation

APPROVED

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I ABSTRACT

This profile is a teaching example. It shows how a single YAML configuration drives three distinct kinds of statement — a **requirement** (SHALL), a **recommendation** (SHOULD) and a **permission** (MAY) — and how the profile builder renders each one differently and treats only requirements as conformance-tested.

Each statement below is labelled by the builder according to one field, `level`. That single field determines the Metanorma block type, the identifier namespace (`/req/`, `/rec/` or `/per/`), the generated filename (`REQ_`, `REC_` or `PER_`) and — most importantly — whether the statement is checked by the Abstract Test Suite in Annex A.

II KEYWORDS

The following are keywords to be used by search engines and document catalogues.

RFC 2119, requirement, recommendation, permission, conformance, OGC API - EDR, service profile

PREFACE

This document was prepared by NOAA/NWS/MDL.

IV SECURITY CONSIDERATIONS

No security considerations have been made for this document.

V SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

- NOAA/NWS/MDL

VI SUBMITTERS

All questions regarding this document should be directed to the editor or the submitters:

Table — Submitters

NAME	AFFILIATION
Shane Mill *(editor)*	NOAA/NWS/MDL

1 SCOPE

1 SCOPE

This standard defines the Statement Levels Explained: Requirement, Recommendation, Permission. It specifies requirements and conformance tests for implementations of this profile.

2

CONFORMANCE

2 CONFORMANCE

Conformance with this standard shall be checked using the Abstract Test Suite in Annex A.

The following conformance classes are defined:

- http://www.opengis.net/spec/ogcapi-edr-3/1.0/conf/levels_explained

3 NORMATIVE REFERENCES

3 NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

Mark Burgoyne, David Blodgett, Charles Heazel, Chris Little, Open Geospatial Consortium: OGC 19-086r6, *OGC API — Environmental Data Retrieval Standard*. Open Geospatial Consortium (2023). http://www.opengis.net/doc/IS/ogcapi-edr-1/1.1.0.

OGC API — EDR Part 3: Service Profiles (draft)

4 TERMS AND DEFINITIONS

4 TERMS AND DEFINITIONS

No terms and definitions are listed in this document.

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this standard.

This document also uses terms defined in OGC API — EDR Part 1: Core.

5

HOW TO READ THIS DOCUMENT

5 HOW TO READ THIS DOCUMENT

The three statements that follow are identical in every respect except their **level**. Read them together to see what the level controls.

5.1. Requirement (SHALL)

A requirement is normative. It is the only statement level that is conformance-tested: every requirement has a matching abstract test in Annex A, and an implementation that does not meet it is non-conformant. Requirements are rendered as a **Requirement** box and carry a `/req/` identifier.

5.2. Recommendation (SHOULD)

A recommendation is informative best practice. It is published as guidance but is **not** conformance-tested — an implementation that does not follow it is still conformant. Recommendations are rendered as a **Recommendation** box and carry a `/rec/` identifier. They never appear in the Abstract Test Suite.

5.3. Permission (MAY)

A permission is an explicitly allowed option. Like a recommendation it is informative and not conformance-tested. Permissions are rendered as a **Permission** box and carry a `/per/` identifier.

5.4. What this means for conformance

Of the three statements below, only the requirement is listed in the Annex A Abstract Test Suite. The recommendation and permission are reported in the profile's quality summary as informative, but they can never make an implementation non-conformant. The author expresses all of this with one field — `level` — and the tooling does the rest.

6

REQUIREMENTS

6 REQUIREMENTS

NOTE: Conformance with this profile is assessed against its requirements (“shall”) only. Recommendations (“should”) and permissions (“may”) are provided as informative guidance: they are not conformance-tested and an implementation is not made non-conformant by not following them. See the quality-assessment summary emitted by the profile tooling for a count of the recommendations and permissions defined by this profile.

REQUIREMENTS CLASS 1	
IDENTIFIER	`http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/levels_explained`
CONFORMANCE CLASS	Conformance class A.1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/conf/levels_explained`
TARGET-TYPE	Statement Levels Explained: Requirement, Recommendation, Permission Profile Standard
NORMATIVE STATEMENTS	Permission 1: `/per/levels_explained/support-post` Recommendation 1: `/rec/levels_explained/document-units` Requirement 1: `/req/levels_explained/output-geojson`

REQUIREMENT 1	
IDENTIFIER	`/req/levels_explained/output-geojson`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/levels_explained`
STATEMENT	A conforming service SHALL return GeoJSON for the example collection. (This is a requirement — it is conformance-tested via Annex A.)
A	The service SHALL expose /collections/example/items.
B	The response Content-Type SHALL be application/geo+json.

RECOMMENDATION 1	
IDENTIFIER	`/rec/levels_explained/document-units`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/levels_explained`

RECOMMENDATION 1

STATEMENT	A conforming service SHOULD document the unit of each parameter. (This is a recommendation — it is informative guidance and is NOT conformance-tested.)
A	Each parameter SHOULD expose a human-readable unit.label.

PERMISSION 1

IDENTIFIER	`/per/levels_explained/support-post`
INCLUDED IN	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/levels_explained`
STATEMENT	A conforming service MAY support POST queries in addition to GET. (This is a permission — it describes an allowed option and is NOT conformance-tested.)
A	Implementations MAY offer POST alongside GET on data query endpoints.

A

ANNEX A (NORMATIVE) ABSTRACT TEST SUITE

A ANNEX A (NORMATIVE) ABSTRACT TEST SUITE

CONFORMANCE CLASS A.1	
IDENTIFIER	`http://www.opengis.net/spec/ogcapi-edr-3/1.0/conf/levels_explained`
REQUIREMENTS CLASS	Requirements class 1: `http://www.opengis.net/spec/ogcapi-edr-3/1.0/req/levels_explained`
CONFORMANCE TEST	Abstract test A.1-1: `/conf/levels_explained/output-geojson`

Example

label	/conf/levels_explained/output-geojson
subject	/req/levels_explained/output-geojson
test-purpose	Validate that the service returns GeoJSON for the example collection.

Send a GET request to /collections/example/items.

Verify the response HTTP status is 200.

Verify the response Content-Type is application/geo+json.